

Bach Phan-Tat

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Computational linguist and data scientist developing interpretable methods for detecting and analysing CHANGE. My current research focuses on lexical and conceptual change using NLP techniques and (diachronic) corpus data. I am also extending this work towards financial time-series analysis and potentially other domains that CHANGE.

WORK EXPERIENCE

The Digital Humanities Institute, University of Sheffield **Sep 2026 – Nov 2026**
Visiting Research Scholar *Sheffield, England*

Vbee (speech technology and conversational AI company) **Feb 2023 – Aug 2024**
Junior Research Engineer *Hanoi, Vietnam*
Project: HUST Chatbot (a conversational AI for Hanoi University of Science and Technology)

- Developed and evaluated machine-learning and BERT-based models for chat domain classification, toxicity detection, and name-based classification, achieving over 85% accuracy on held-out evaluation sets.
- Designed data-sampling and human-evaluation procedures, applying statistical methods to assess dataset quality, annotation reliability, and model performance.
- Built an LLM-assisted chatbot evaluation pipeline, including evaluation criteria, prompt design, automated response scoring, and systematic analysis of model outputs.
- Developed language-data pipelines including web crawling, information extraction, LLM-assisted data generation, preprocessing, annotation, and quality control for text and speech datasets.
- Contributed to the development of linguistic resources, and rule-based components for speech and language tasks, including speech-to-text, text-to-speech, and part-of-speech parsing.

what3words (location-mapping technology company) **Jul 2023 – Aug 2023**
Remote Research Consultant *London, England*
Project: Vietnamese Automatic Speech Recognition system (in collaboration with VinFast)

- Conducted systematic linguistic error analysis of Vietnamese ASR outputs, categorising recurring errors involving phonetic confusions, lexical ambiguity, and contextual misrecognition.
- Translated error-analysis findings into data- and model-level recommendations.

ACADEMIC & PROFESSIONAL SERVICE

Supervising and Mentoring

- Advised a Master's thesis in Linguistics at Lancaster University, which is awarded a Distinction (2024).
- Mentored applicants in postgraduate and fellowship applications, with subsequent admissions to the Linguistics PhD programme at Lancaster University (2024), the YSEALI Fellowship (2024), the Stipendium Hungaricum Scholarship (2026).

Reviewing

- Reviewer for Cognitive Linguistic Studies (2026), Linguists' Day, Linguistic Society of Belgium (2025).

Organising

- Co-organised the PhD Day for the Department of Linguistics at KU Leuven. 2026.

PUBLICATIONS

Book chapters (peer-reviewed)

- Sofía Aguilar-Valdez*, **Bach Phan-Tat***, Dirk Speelman, Dirk Geeraerts, and Stefania Degaetano-Ortlieb. 2026. Discursive parallels of the chemical revolution: topic modelling and distributional analysis. In *Understanding Science with Large Language Models? Potentials for the History, Philosophy, and Sociology of Science* (open access, transcript). (Book chapter / edited volume). *Equal contribution.

Conference and workshop proceedings (peer-reviewed)

- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, and Dirk Speelman. 2026. ReFRAME or Remain: Unsupervised Lexical Semantic Change Detection with Frame Semantics. *Proceedings of the 15th Joint Conference on Lexical and Computational Semantics (*SEM 2026)*, *ACL 2026*.
- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, and Dirk Speelman. 2026. Transparent Semantic Change Detection with Dependency-Based Profiles. *Proceedings of the 6th International Workshop on Computational Approaches to Historical Language Change (LChange'26)*, *EACL 2026*.
- Khai Le-Duc, Tuyen Tran, **Bach Phan-Tat**, et al. 2025. MultiMed-ST: Large-scale Many-to-many Multilingual Medical Speech Translation. *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025)*.
- Khai Le-Duc, Phuc Phan, Tan-Hanh Pham, **Bach Phan-Tat**, Minh-Huong Ngo, Thanh Nguyen-Tang, Truong-Son Hy. 2025. MultiMed: Multilingual Medical Speech Recognition via Attention Encoder Decoder. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, Industry Track (ACL 2025)*. Top 15% of accepted papers.
- Khai Le-Duc, Khai-Nguyen Nguyen, **Bach Phan-Tat**, Le Duy, Long Vo-Dang, Truong-Son Hy. 2025. Sentiment Reasoning for Healthcare. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, Industry Track (ACL 2025)*. Oral presentation.

Journal articles (peer-reviewed)

- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, Dirk Speelman. 2026. Evaluating the Evaluator: Problems with SemEval-2020 Task 1 for Lexical Semantic Change Detection. *Journal of Open Humanities Data*.
- Catherine Wong, **Bach Phan-Tat**, Susan Fitzmaurice. 2026. Methods, Data, and Conceptual Change: Reflections from Two Quantitative Diachronic Case Studies. *Journal of Open Humanities Data*.

Pre-prints and under-review articles

- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, Dirk Speelman. 2026. SynFlow: A Multidimensional Diachronic Semantic Analysis Toolkit. Preprint
- Duc Cao-Dinh, Khai Le-Duc, Anh Dao, **Bach Phan-Tat**, Chris Ngo, Duy M. H. Nguyen, Nguyen X. Khanh, Thanh Nguyen-Tang. 2025. Audio-3DVG: Unified Audio - Point Cloud Fusion for 3D Visual Grounding. (Preprint).

PRESENTATIONS

Conference presentations

- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, and Dirk Speelman. 2026. Underlying Dimensions of Semantic Change. Digital Humanities Congress 2026.
- **Bach Phan-Tat**, Kris Heylen, and Stefano De Pascale. 2026. Lexical Semantic Change Detection? A Good Parser Might Be Enough! 1st LOT Conference.
- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, and Dirk Speelman. 2026. SynFlow: Continuous Semantic Change Analysis via Dependency Co-occurrences. ICAME 47.
- **Bach Phan-Tat**, Dirk Geeraerts, and Dirk Speelman. 2026. Conceptual Change during the Chemical Revolution: Air, Acid, and Water in the Royal Society Corpus. ICAME 47.
- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, and Dirk Speelman. 2026. From Parsers to Prompts: Combining AI and Linguistics for Interpretable Language Evolution. AI in Language Evolution Workshop, Evolang 2026.
- **Bach Phan-Tat**. 2026. Slot-Filler Distributional Changes in Times of Scientific Debate: A Case Study of Air. Quantitative Diachronic Linguistics and Cultural Analytics.
- Khai Le-Duc, Khai-Nguyen Nguyen, **Bach Phan-Tat**, et al. 2024. Sentiment Reasoning for Healthcare. Advancements In Medical Foundation Models: Explainability, Robustness, Security, and Beyond Workshop, NeurIPS 2024.

Invited talks & seminars

- **Bach Phan-Tat**. 2026. Underlying Dimensions of Semantic Change. NLP & text mining seminar organised by Leuven.AI.
- **Bach Phan-Tat**. 2026. SynFlow: Continuous Semantics Change Analysis via Dependency Co-occurrences. Data in Historical Linguistics seminar series.

EDUCATION

KU Leuven

PhD, Computational Linguistics

Thesis: **Underlying Dimensions in Conceptual Change**

Sep 2024 – Present

Leuven, Belgium

Selected Achievements

- Developed a lightweight and interpretable framework for lexical semantic change detection and analysis based on structured linguistic patterns and diachronic distribution comparison.
- Developed SynFlow, an efficient Python package for extracting syntactic co-occurrences and constructions, comparing their distributions across time, and incrementally clustering diachronic static embeddings.
- Built automated pipelines for LLM-assisted linguistic annotation; and for cleaning, lemmatising, and parsing large corpora.
- Optimised computational workflows through parallel and batch processing, reducing processing time by 80%.
- Maintained NephoSem, a lexical-semantics toolkit developed by the QLVL research unit, and adding support for other types of static embedding.

Thesis supervisors: Prof. Dr. Dirk Speelman, Prof. Dr. Dirk Geeraerts, Dr. Kris Heylen

WorldQuant University

MSc, Financial Engineering

Finished modules: Financial Market, Financial Data, Financial Econometrics

Oct 2025 – Present

Remote

University of Stirling

Sep 2021 – Sep 2022

Thesis: **The structure of *thời gian*: A semantic analysis of the Vietnamese lexeme *thời gian* based on Principled Polysemy**

Selected Achievements

- Crawled, cleaned, and compiled a specialised corpus of 500 randomly sampled Vietnamese news articles.
- Identified seven senses of the Vietnamese lexeme *thời gian* ('time'), their semantic structure and semantic development using the Principled Polysemy framework.

Thesis supervisor: Dr. Andrew Smith

Thesis grade: 80% (Distinction; class rank: 1st)

Average grade: 78% (Distinction; class rank: 1st)

Academy of Finance

BA, Corporate Finance

Aug 2016 – Aug 2020

Hanoi, Vietnam

Relevant modules: Linear Algebra, Calculus, Probability and Statistics, Econometrics

TECHNICAL SKILLS

Programming & Computing

- Programming languages (ordered by proficiency): Python (NumPy, Pandas, Matplotlib, scikit-learn, spaCy, PyTorch, Hugging Face), R, Bash, LaTeX
- Parallel, batch, and GPU-enabled computing workflows
- Databases: MongoDB, SQL
- Git/GitHub

Data Science

- Exploratory data analysis, data preprocessing and (interactive) data visualisation
- Machine learning and deep learning for NLP, time-series, structured data and computer vision
- Quantitative research design, experimental design, and A/B testing
- LLM-assisted annotation and evaluation workflows
- Mathematics: linear algebra, multivariate calculus, statistics

Linguistics & Corpus Methods

- Corpus tools: CQPweb, #LancsBox, Sketch Engine, AntConc
- Linguistic analysis: phonetics, morphology, syntax, semantics
- Linguistic annotation, transcription, and corpus preparation

AWARDS AND HONOURS

- **2026: First Place, Databricks Challenge** (US\$1,000), Hack-Nation's 4th Global AI Hackathon; 3,000+ participants from 300+ universities across 65+ countries.
- **2024: Marie Skłodowska-Curie Actions Doctoral Fellowship** (fully funded for three years), Horizon Europe; awarded to 10 out of 284 applicants worldwide.
- **2022: Research-Based Learning Prize for the best Master's dissertation in Literature and Languages** (GBP 225), University of Stirling; the first recipient in the history of the Linguistics department.
- **2021: Postgraduate Scholarship** (GBP 4,000), University of Stirling.

SELECTED ADDITIONAL TRAINING

Data Science

- Inferential Statistics (The University of Amsterdam)
- Applied Data Science Lab (WorldQuant University)
- Deep Learning (Neuromatch Academy)
- Applied AI Lab: Deep Learning for Computer Vision (WorldQuant University)

Linguistics

- Corpus Linguistics: Method, Analysis, Interpretation (Lancaster University)
- Corpus Linguistics Summer School (Birmingham University)

Financial Engineering

- Foundations of Financial Engineering (WorldQuant University)

Research Methodology

- Quantitative Method (The University of Amsterdam)
- Qualitative Research Methods (The University of Amsterdam)

LANGUAGES

- Vietnamese: Native
- English: IELTS 8.5 – C2
- Dutch: A2